Practical: 9 Data-Driven Application

Develop a data driven user authentication application that asks for user id and password to user. After accepting user id and password, application should check whether entered authentication details exists in data table or not. Display appropriate message on screen

Create an application that:

- Takes **User ID** and **Password**
- Stores user data in SQLite
- Checks whether the entered User ID and Password exist
- Displays **Login Successful** or **Invalid User**

Step 1: Create New Project

- Open Android Studio
- Click **New Project**
- Select **Empty Views Activity**
- Language: **Java**
- Finish

Step 2: Create DatabaseHelper.java

Right Click package → **New** → **Java Class**

Name: **DatabaseHelper**

```
package com.example.authenticationapp;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
```

```
public class DatabaseHelper extends SQLiteOpenHelper {

    public DatabaseHelper(Context context) {
        super(context, "UserDB", null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {

        db.execSQL("CREATE TABLE users(userid TEXT PRIMARY KEY,password TEXT)");

        //Default User
        db.execSQL("INSERT INTO users VALUES('admin','1234')");
        db.execSQL("INSERT INTO users VALUES('student','1111')");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

        db.execSQL("DROP TABLE IF EXISTS users");
        onCreate(db);
    }

    //Check Login

    public boolean checkUser(String userid,String password){

        SQLiteDatabase db=this.getReadableDatabase();

        Cursor cursor=db.rawQuery("SELECT * FROM users WHERE userid=? AND
password=?",
                new String[]{userid,password});

        if(cursor.getCount()>0){
            return true;
        }
        else{
            return false;
        }
    }
}
```

Step 3: activity_main.xml

Open

res
  layout
    activity_main.xml

Replace code with

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etUser"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter User ID"/>

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"/>

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"/>

</LinearLayout>
```

Step 4: MainActivity.java

Replace code

```
package com.example.authenticationapp;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    EditText etUser,etPassword;
    Button btnLogin;

    DatabaseHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etUser=findViewById(R.id.etUser);
        etPassword=findViewById(R.id.etPassword);
        btnLogin=findViewById(R.id.btnLogin);

        db=new DatabaseHelper(this);

        btnLogin.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {

                String user=etUser.getText().toString();
                String pass=etPassword.getText().toString();

                if(db.checkUser(user,pass))
                {
                    Toast.makeText(MainActivity.this,
                            "Login Successful",
```

```
                    Toast.LENGTH_SHORT).show();
            }
            else
            {
                Toast.makeText(MainActivity.this,
                        "Invalid User ID or Password",
                        Toast.LENGTH_SHORT).show();
            }

        }
    });

  }
}
```

Step 5: Run Application

The database is automatically created with two users.

User ID	Password
admin	1234
student	1111

Output

Case 1

Enter

User ID : admin
Password : 1234

Output

Login Successful

Case 2

Enter

User ID : abc
Password : 123

Output

Invalid User ID or Password